

SAMUEL ROTH

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EDUCATION:

Syracuse University, College of Engineering and Computer Science, Syracuse, NY

Expected May 2027

B.S. Aerospace Engineering, Minor in Engineering Management

GPA: 3.7

Honors: Sigma Gamma Tau National Aerospace Engineering Honors Society, Dean's List Fall 2023 - Present, National Honors Fraternity Phi Sigma Pi, Jefferson Award for Public Service

SKILLS & INTERESTS:

Engineering: ANSYS Mechanical, ANSYS Fluent, Cameo Systems Modeler, SolidWorks, Creo, Python

Industry: Solumina, SAP, MATLAB, Tungsten Power PDF, Adobe Acrobat, Windows & Google Suite

Certifications: MOTIS 622i Flight Simulator Operator, FAA Recreational UAS Safety, Wilderness First Aid

WORK EXPERIENCE:

Avionics Engineering Intern, Lockheed Martin RMS (Sikorsky), Stratford CT

May 2026 – Present

- Supported avionics integration and production for military helicopter electrical systems by reviewing and redlining electrical schematics against ICDs, engineering specifications, and manufacturing requirements.
- Reviewed aircraft wiring diagrams, cable assemblies, connector and backshell configurations, and coaxial cable installations to identify discrepancies and support engineering and manufacturing teams.
- Developed process improvements to increase production accuracy and efficiency, including evaluating RFID-based part identification systems for hundreds of hardware storage locations.
- Achieved 1st place among 10 intern teams in Sikorsky's Engineering & Technology competition by developing a conceptual hybrid-electric propulsion system; translated SOW requirements into trade studies, architecture decisions, and thermal/structural analyses to validate the selected design.
- Continued with Lockheed Martin part-time during the academic year following the summer internship, supporting model-based systems engineering (MBSE) in Cameo and avionics system development for project creation and development.

Engineering Living Learning Community Resident Advisor, Syracuse University, Syracuse NY

August 2024 – May 2026

- Managed 38 residents within a 100+ person living-learning community, facilitating roommate agreements, resolving conflicts, and serving on-call for crisis response during overnight and weekend shifts.
- Promoted community engagement by organizing large-scale events such as a pumpkin patch trip with over 70 participants, while also building connections through door-to-door check-ins and academic support.

Aviation Camp Instructor, Hiller Aviation Institute, San Carlos CA

May 2025 – August 2025

- Taught aviation fundamentals and STEM concepts to 20+ students ages 7–12 through hands-on activities and educational field trips to the USS Hornet and airport observation areas.

ENGINEERING APPLICATIONS:

Airfoil Flow & Lift Simulator, Syracuse University, Syracuse NY

January 2026 – April 2026

- Developed a 1,000-panel MATLAB airfoil simulator for a NACA 4412 profile to predict pressure coefficient, velocity fields, streamlines, circulation, and lift across 0°, 10°, and 20° angles of attack.
- Implemented a point-vortex circulation model and Kutta condition to simulate realistic lifting flow and trailing-edge behavior, producing lift coefficients of 0.519, 1.714, and 2.857 and evaluating the effects of camber and angle of attack.
- Validated results against thin airfoil theory and XFOIL, achieving lift predictions within approximately 6.5% of inviscid XFOIL at 10° and analyzing how Reynolds number and viscosity affect pressure distribution, lift, and drag.

Wind Tunnel Aerodynamic Laboratory, Syracuse University, Syracuse NY

August 2025 – December 2025

- Designed and executed wind tunnel experiments to measure flow velocity, vortex shedding, wake behavior, boundary layers, and aerodynamic forces on airfoils and finite-length cylinders.
- Developed MATLAB analysis for pitot tube calibration, FFT-based frequency analysis, drag coefficient calculation, and uncertainty calculation, achieving a calibration fit with $R^2 = 0.9988$ and reduced uncertainty at higher tunnel speeds.
- Compared experimental results to theoretical predictions in MATLAB including Bernoulli flow assumptions, Strouhal relations, and published cylinder drag data, confirming near-constant Strouhal number behavior and drag coefficients within approximately ± 0.3 of expected values.

EXTRACURRICULAR ACTIVITIES:

President, Syracuse Chapter of the American Institute Aeronautics and Astronautics (AIAA)

August 2023 – Present

- Lead the Syracuse AIAA chapter and executive board in organizing technical programming, professional development, industry engagement, peer tutoring initiatives, and conference participation for aerospace engineering students.
- Coordinate faculty and industry events while expanding student involvement in AIAA programs, including preparations to send approximately 10 students to the Region I Student Conference and support technical paper presentations.

Treasurer & Secretary, ORBiT: Syracuse Rocketry Club

October 2023 – Present

- Manage budgeting, purchasing, financial planning, and organization administration for a 90+ member collegiate rocketry club while leading technical workshops, rocket builds, launches, and preparation for NAR Level 1 High-Power Rocketry certification.
- Support development of a rocket propulsion test stand by programming a microcontroller-based thermocouple data acquisition system and integrating sensors, breadboard circuitry, and instrumentation for experimental nozzle testing.